

Monday, Apr 8

Probability Density, Survival, and Hazard Functions

Let T be a continuous random variable that is time-till-event. Four related functions are used to describe the distribution of T .

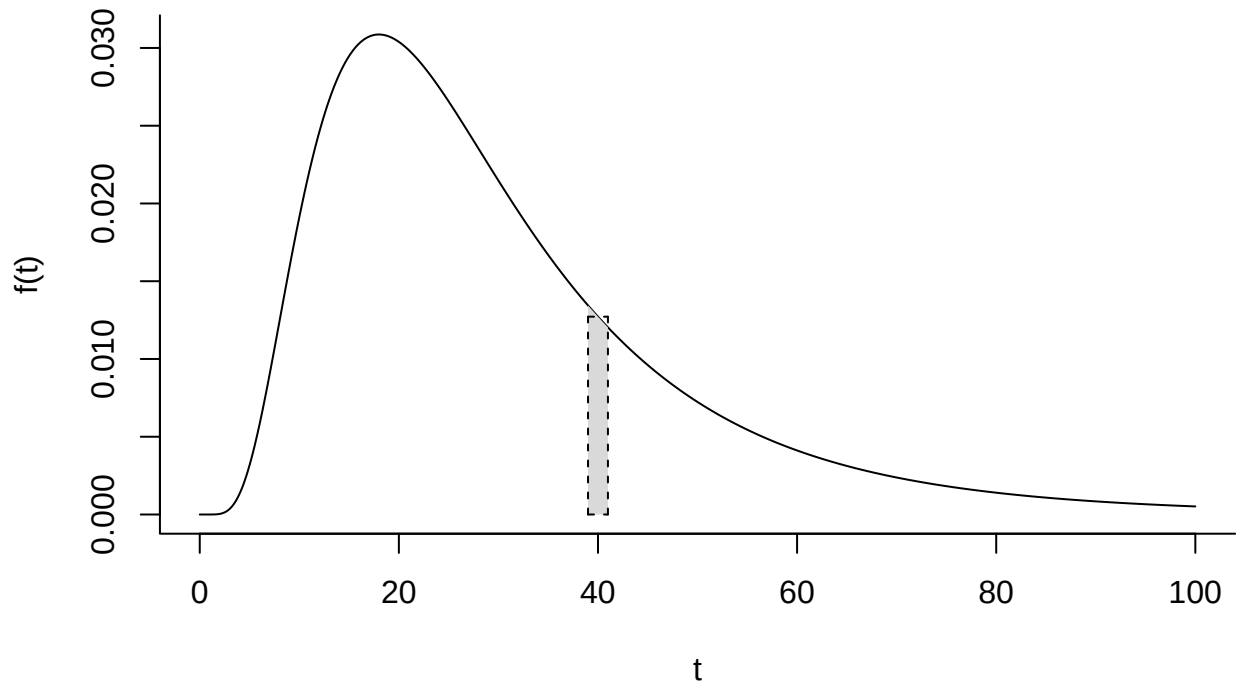
The Probability Density Function

The *probability density function* of T is

$$f(t) = \lim_{\delta t \rightarrow 0} \frac{P(t \leq T < t + \delta t)}{\delta t}.$$

If δ is relatively small then $P(t \leq T < t + \delta t) \approx f(t)\delta t$ and so $f(t) \approx P(t \leq T < t + \delta t)/(\delta t)$ and thus $f(t)$ is approximately *proportional* to the probability that T is between t and $t + \delta t$. So $f(t)$ is approximately *proportional to the probability that the event will happen “near” t* .

For the distribution below, the probability that T is *approximately* 40 (say, between 39 and 41) equals the area under the curve and between 39 and 41. This probability is *approximated* by the rectangle, which has area $wf(40)$, where $w = 2$ is the width of the rectangle and $f(40)$ is the height of the rectangle. So the probability that T is approximately 40 is *proportional to* $f(40)$.

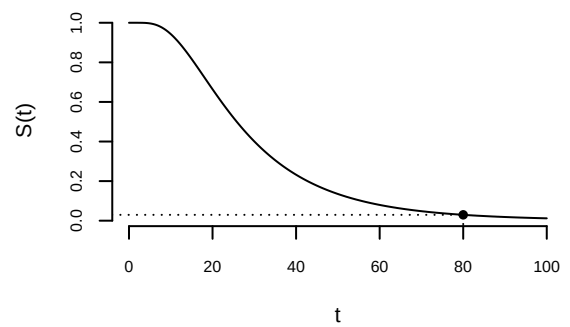
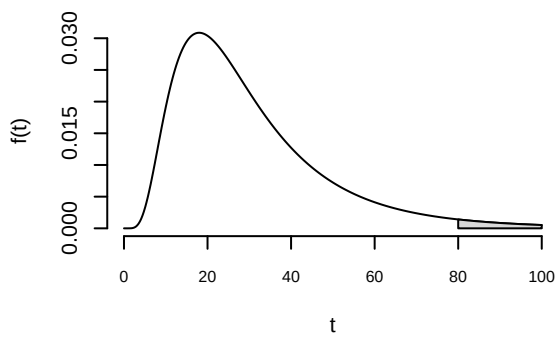
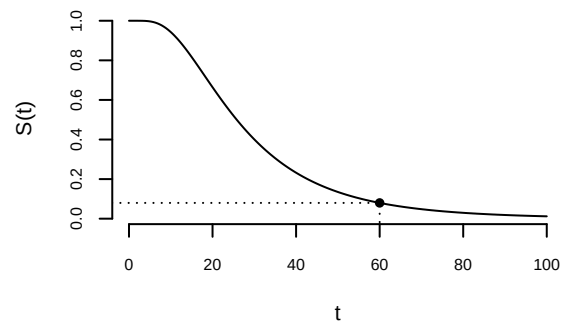
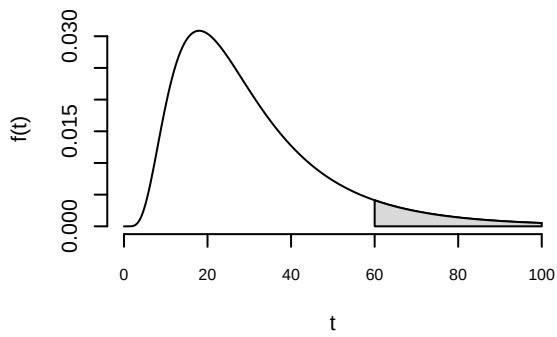
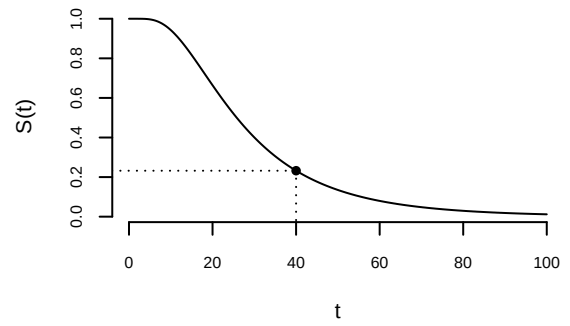
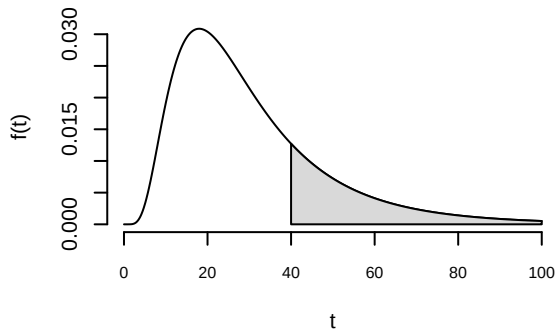
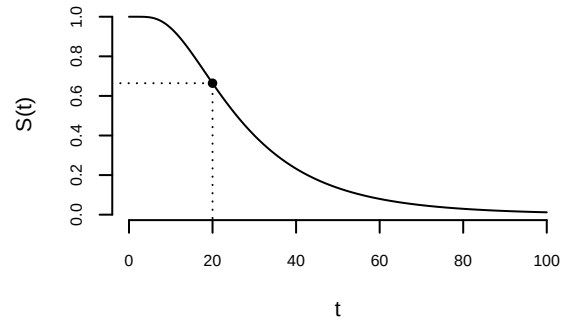
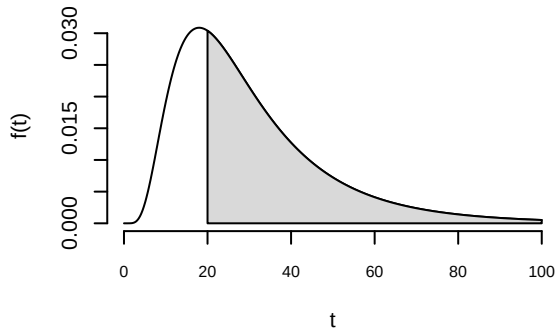


The Survival Function

The *survival function* is

$$S(t) = P(T \geq t).$$

It equals the area under $f(t)$ and between t and ∞ . The area under $S(t)$ equals $E(T)$ if $S(0) = 1$ and $S(\infty) = 0$.



Thus

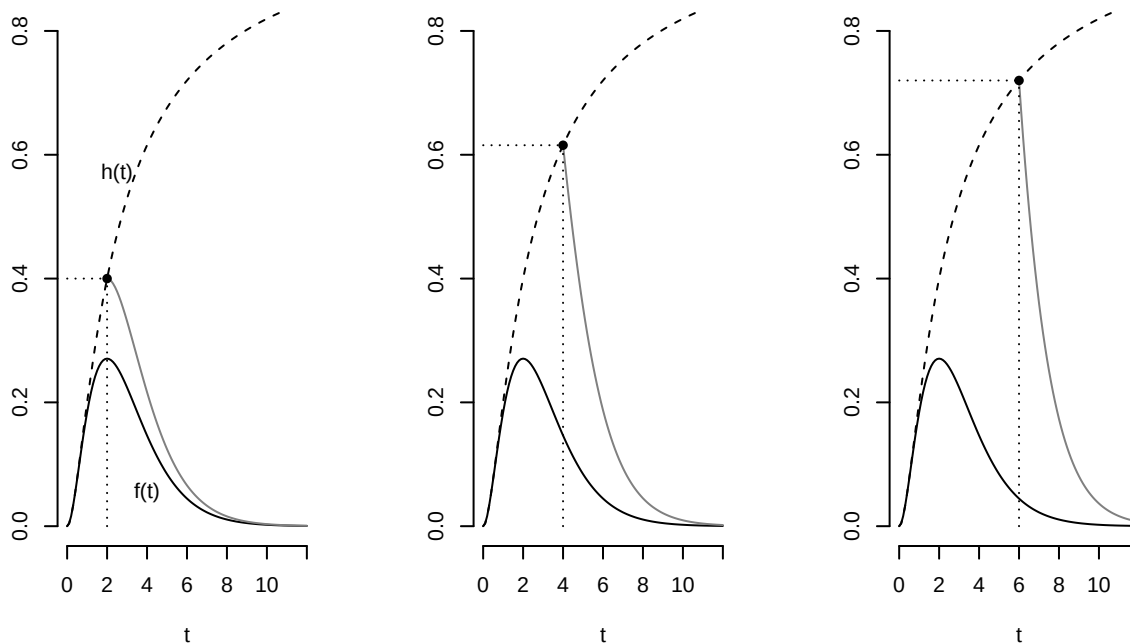
$$S(t) = \int_t^{\infty} f(z) dz.$$

The Hazard Function

The *hazard function* is

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{P(t \leq T < t + \delta t | T \geq t)}{\delta t} = \frac{f(t)}{S(t)}.$$

If δ is relatively small then $h(t)$ is approximately *proportional* to the probability that $t \leq T < t + \delta t$ given survival up to t — i.e., $T \geq t$. So $h(t)$ is approximately proportional to the probability of the event happening at near time t if it has not yet happened.



Distributions and Hazard Functions

A wide variety of distributions can be used for parametric survival models such as AFT models. Below is a list of just some of those distributions. One of the more noticeable differences between them is the shape of their hazard functions.

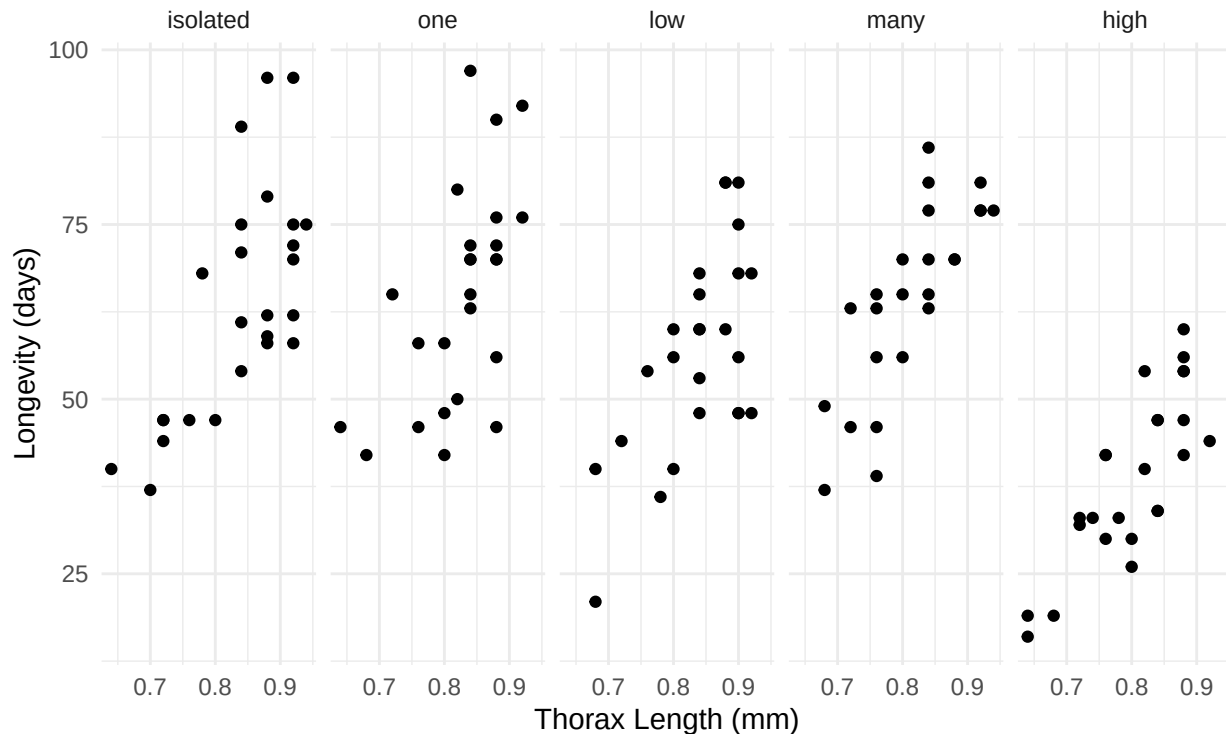
1. *Log-normal*. The distribution of $\log(T_i)$ is normal. Single-peaked hazard function. Known as `lognormal` by `survreg` and `flexsurvreg`, and also `lnorm` by `flexsurvreg`.
2. *Log-logistic*. The distribution of $\log(T_i)$ is logistic. Single-peaked or decreasing hazard function. Known as `loglogistic` by `survreg` and `flexsurvreg` and `llogis` by `flexsurvreg`.
3. *Gamma*. Monotonic or flat hazard function. Known as `gamma` to `flexsurvreg`.
4. *Weibull*. Monotonic or flat hazard function. Known as `weibull` to both `survreg` and `flexsurvreg`.
5. *Exponential*. Flat hazard function (“memoryless”). Known as `exp` to `flexsurvreg` but also as a special case of `weibull` if `scale = 1` with `survreg`.
6. *Gompertz*. Increasing hazard function. Known as `gompertz` to `flexsurvreg`.
7. *Generalized gamma*. Monotonic, single-peaked, and “bathtub” hazard functions. The exponential, Weibull, gamma, and log-normal are special cases. Known as `gengamma` to `flexsurvreg`.
8. *Generalized F*. Single-peaked or decreasing. Known as `genf` to `flexsurvreg`.

Estimating and Plotting Hazard Functions

The `summary` function can be used to estimate the hazard function based on a `flexsurvreg` model object.

Example: Consider data from an experiment on the effects of sexual activity on the lifespan of the male fruitfly. Thorax length was used as a covariate.

```
library(faraway)
p <- ggplot(fruitfly, aes(x = thorax, y = longevity)) +
  geom_point() + facet_wrap(~ activity, ncol = 5) +
  labs(x = "Thorax Length (mm)", y = "Longevity (days)") +
  theme_minimal()
plot(p)
```

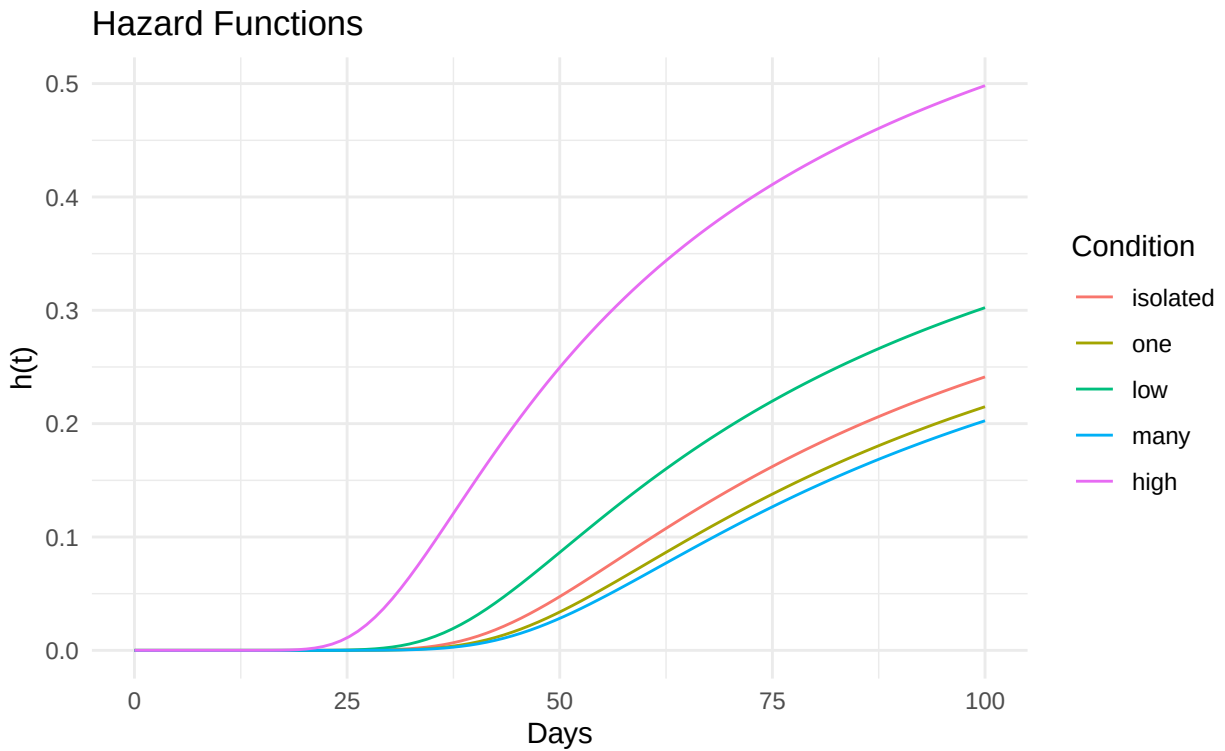


```
m <- flexsurvreg(Surv(longevity) ~ activity + thorax,
  data = fruitfly, dist = "gamma")

d <- data.frame(activity = levels(fruitfly$activity), thorax = 0.8)
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "hazard", tidy = TRUE)
head(d)
```

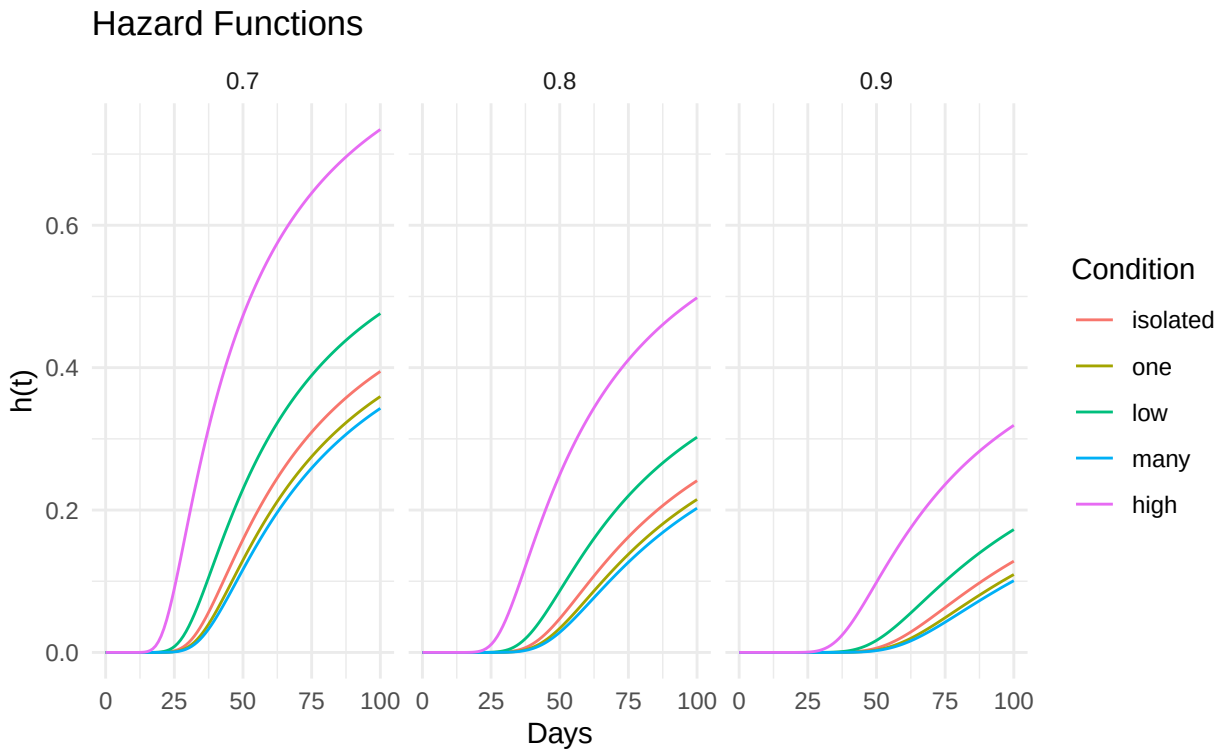
	time	est	lcl	ucl	activity	thorax
1	0.000	0.000e+00	0.000e+00	0.000e+00	isolated	0.8
2	1.010	1.269e-38	1.180e-48	1.837e-30	isolated	0.8
3	2.020	1.901e-30	3.094e-38	3.849e-24	isolated	0.8
4	3.030	9.297e-26	2.246e-32	1.804e-20	isolated	0.8
5	4.040	1.701e-22	3.038e-28	6.760e-18	isolated	0.8
6	5.051	5.138e-20	4.025e-25	6.079e-16	isolated	0.8

```
p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "h(t)", color = "Condition", title = "Hazard Functions")
plot(p)
```



```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "hazard", tidy = TRUE)

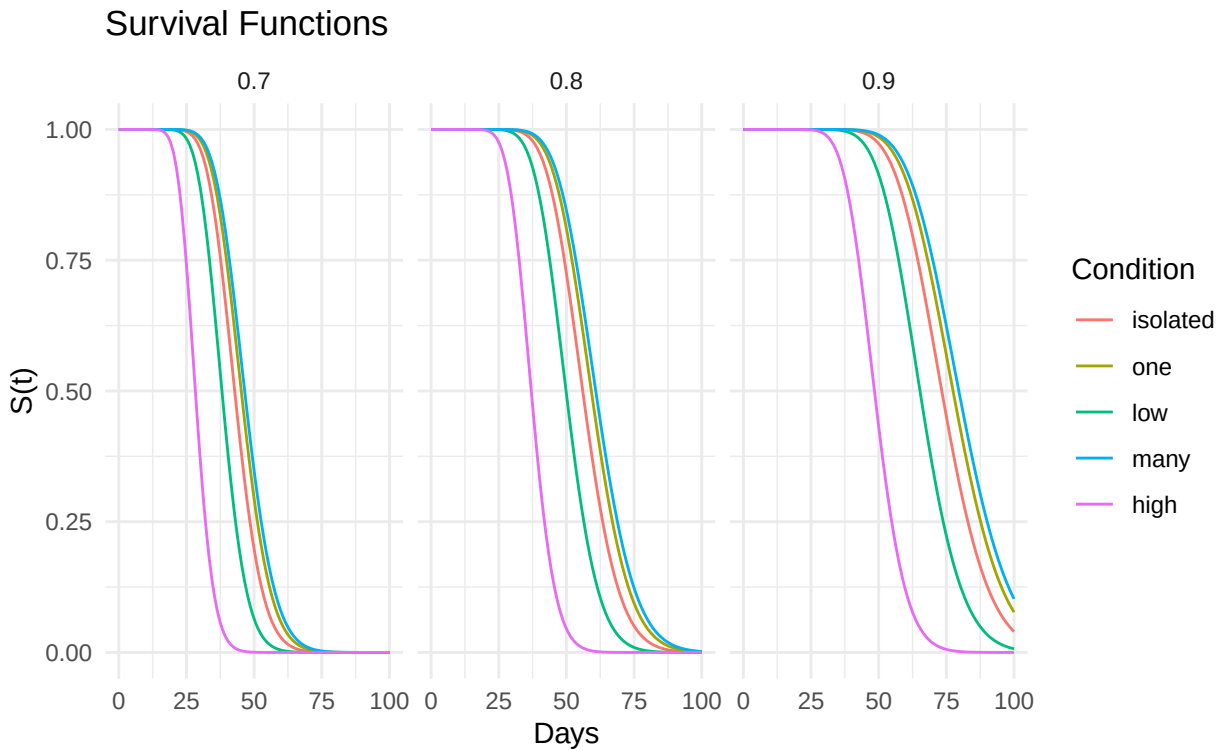
p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "h(t)", color = "Condition", title = "Hazard Functions") +
  facet_wrap(~ thorax)
plot(p)
```



For comparison here are the *survival functions*.

```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "survival", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "S(t)", color = "Condition", title = "Survival Functions") +
  facet_wrap(~ thorax)
plot(p)
```

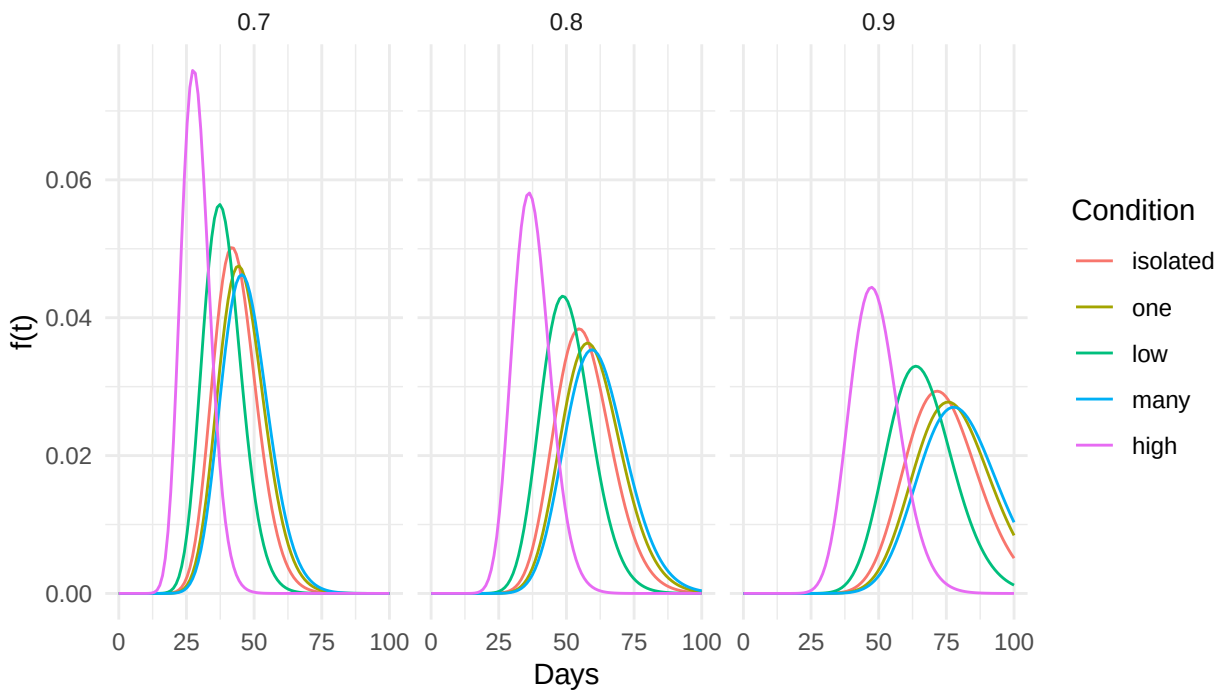


And here are the *probability density functions*.

```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  fn = function(t, ...) dgamma(t, ...), tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "f(t)", color = "Condition", title = "Density Functions") +
  facet_wrap(~ thorax)
plot(p)
```

Density Functions



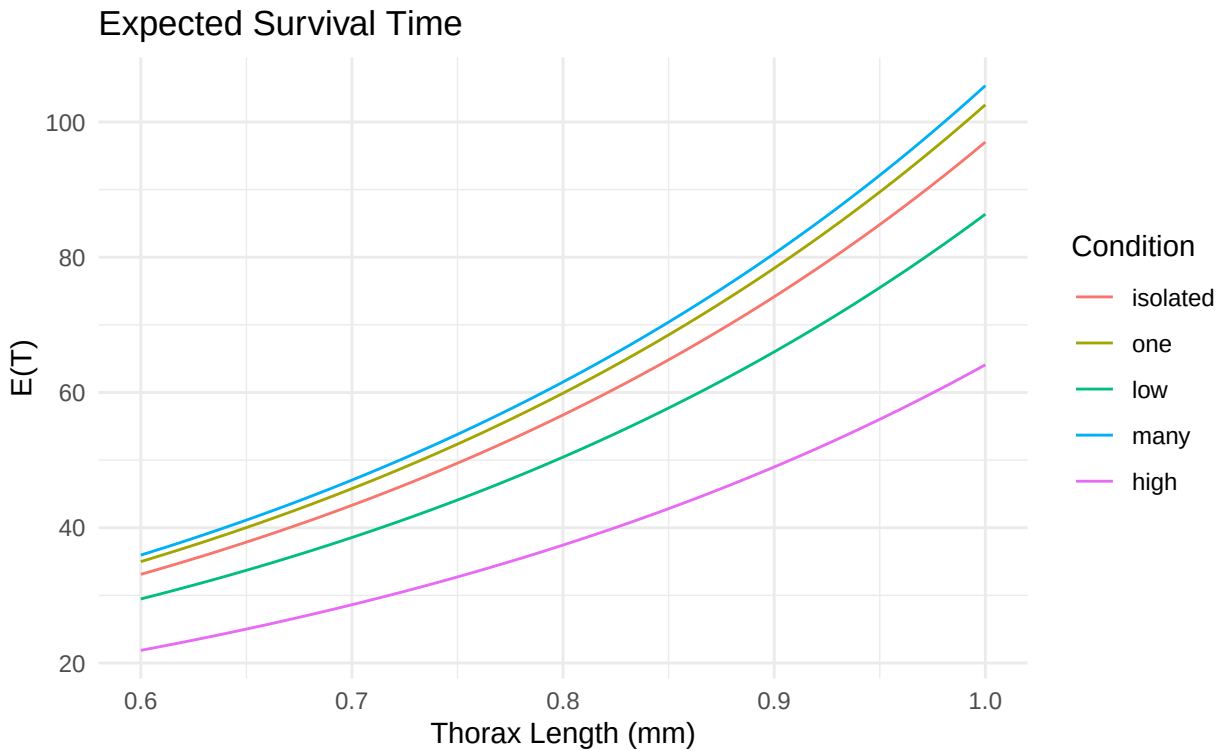
Note that we can adapt this to other distributions by adding a `d` to the beginning of the distribution name recognized by `flexsurvreg`. This include log-normal (`dlnorm`), log-logistic (`dllogis`), gamma (`dgamma`), Weibull (`dweibull`), exponential (`dexp`), Gompertz (`dgomperz`), generalized gamma (`dgengamma`), and generalized F (`dgenf`).

Finally we can also plot the *expected* survival time. This is analogous to using `predict` with `type = response` in a GLM.

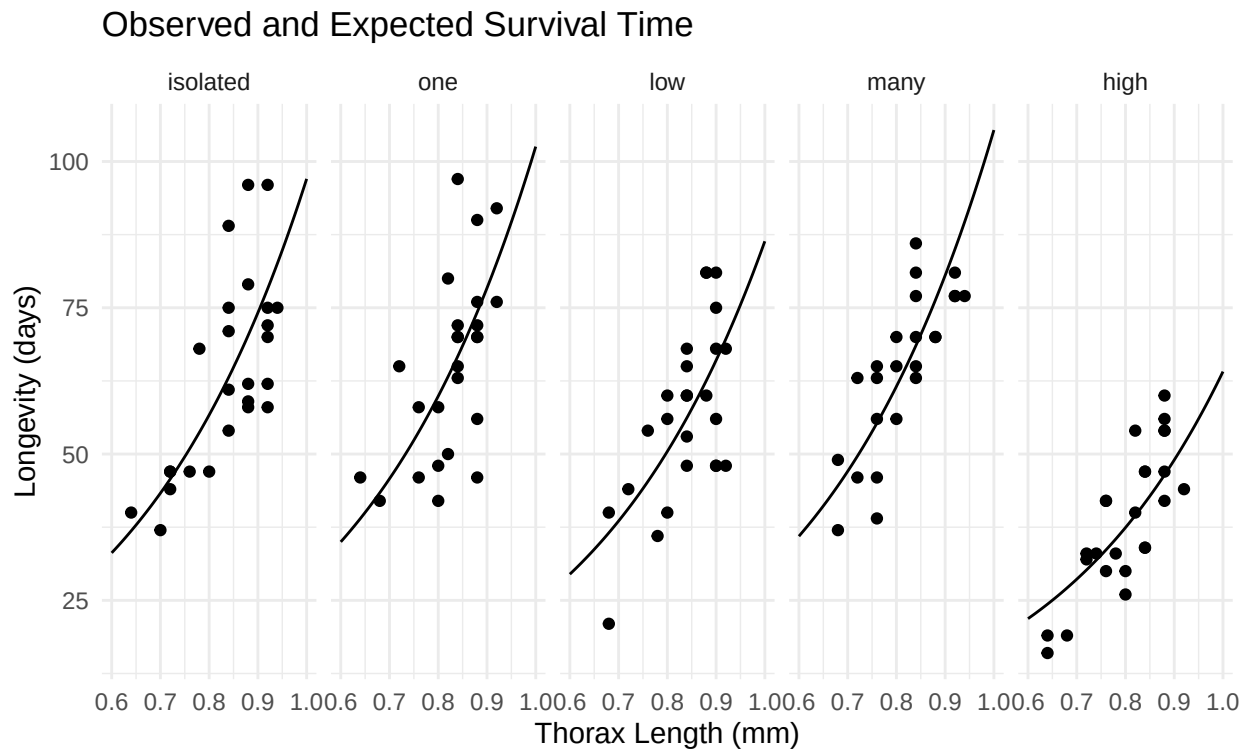
```
d <- expand.grid(activity = levels(fruitfly$activity),
  thorax = seq(0.6, 1.0, length = 100))
d <- summary(m, newdata = d, type = "mean", tidy = TRUE)
head(d)
```

	est	lcl	ucl	activity	thorax
1	33.11	29.10	37.51	isolated	0.600
2	34.99	31.18	39.54	one	0.600
3	29.47	25.94	33.55	low	0.600
4	35.96	31.96	40.49	many	0.600
5	21.87	19.49	24.61	high	0.600
6	33.47	29.45	37.88	isolated	0.604

```
p <- ggplot(d, aes(x = thorax, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Thorax Length (mm)", y = "E(T)", color = "Condition",
    title = "Expected Survival Time")
plot(p)
```

```
p <- ggplot(fruitfly, aes(x = thorax, y = longevity)) +
  geom_point() + facet_wrap(~ activity, ncol = 5) +
  labs(x = "Thorax Length (mm)", y = "Longevity (days)",
       title = "Observed and Expected Survival Time") +
  theme_minimal() + geom_line(aes(y = est), data = d)
plot(p)
```



Example: Consider an AFT model for the leukemia data.

```
library(survival)
leukemia$status <- factor(leukemia$status, labels = c("alive","dead"))

m <- flexsurvreg(Surv(time, status == "dead") ~ x,
  dist = "weibull", data = leukemia)

# create plot of hazard functions
d <- data.frame(x = c("Maintained","Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "hazard", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "h(t)", linetype = "Extended",
    title = "Hazard Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.7, 0.5))
p.h <- p

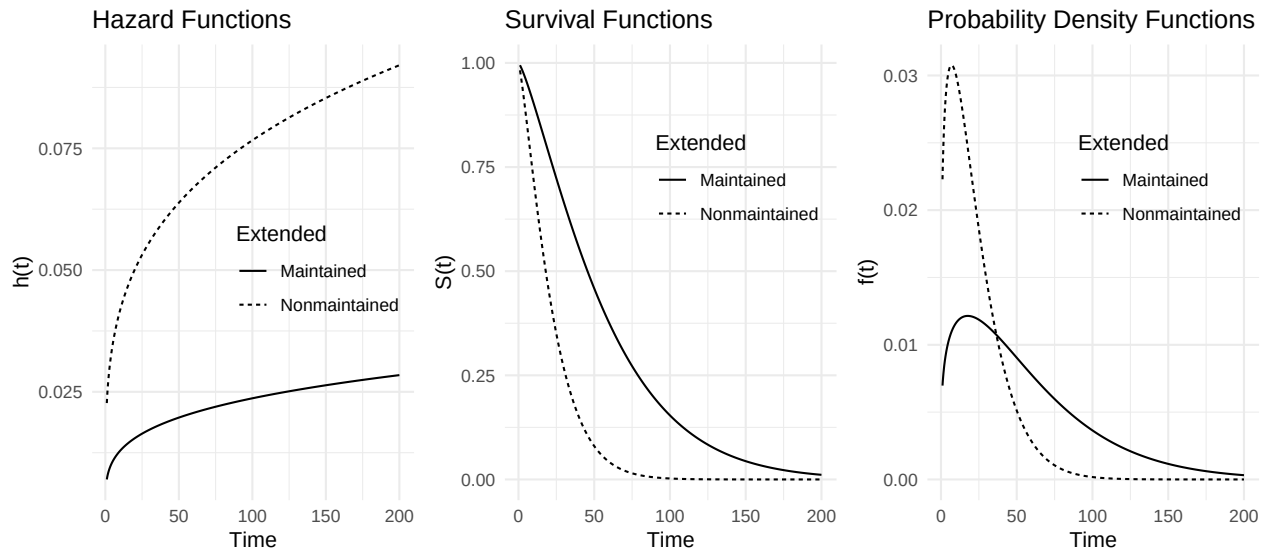
# create plot of survival functions
d <- data.frame(x = c("Maintained","Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "survival", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "S(t)", linetype = "Extended",
    title = "Survival Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.7, 0.7))
p.s <- p

# create plot of probability density functions
d <- data.frame(x = c("Maintained","Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  fn = function(t, ...) dweibull(t, ...), tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "f(t)", linetype = "Extended",
    title = "Probability Density Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.7, 0.7))
p.d <- p

# put the plots together into one plot
cowplot::plot_grid(p.h, p.s, p.d, ncol = 3)
```

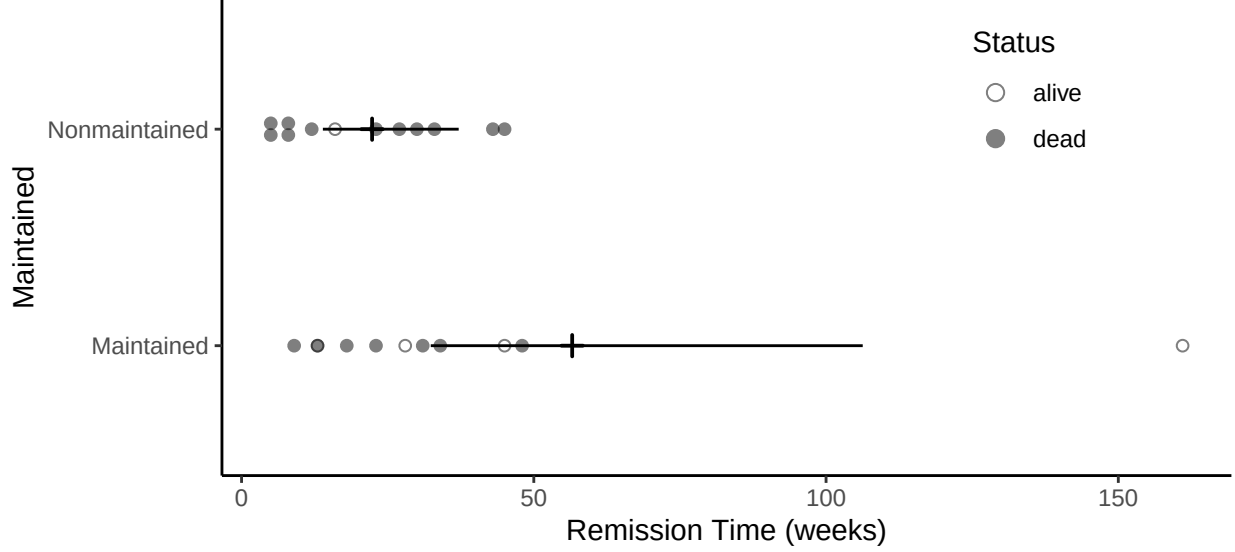


We can also plot the raw data with the estimated expected survival times and confidence intervals for the estimated expected survival time.

```
d <- summary(m, newdata = data.frame(x = c("Maintained", "Nonmaintained")),
  type = "mean", tidy = TRUE)
d
```

```
      est   lcl   ucl      x
1 56.57 32.37 106.27 Maintained
2 22.33 13.91  37.15 Nonmaintained
```

```
p <- ggplot(leukemia, aes(x = x, y = time)) +
  geom_dotplot(aes(fill = status), stackdir = "center", binaxis = "y",
    binwidth = 1, dotsize = 2, alpha = 0.5) + coord_flip() +
  scale_fill_manual(name = "Status", values = c("white", "black")) +
  geom_pointrange(aes(y = est, ymin = lcl, ymax = ucl),
    shape = 3, data = d) +
  labs(x = "Maintained", y = "Remission Time (weeks)") +
  theme_classic() +
  theme(legend.position = "inside", legend.position.inside = c(0.8, 0.8))
plot(p)
```



A very useful feature of the **flexsurv** package is that a user can program their own distribution for use with the functions therein.

Proportional Hazards Models

Let $h_0(t)$ be the “baseline” hazard function (i.e., the hazard function when all $x_j = 0$). A *proportional hazards model* has the form

$$h_i(t) = h_0(t)e^{\beta_1 x_{i1}} e^{\beta_2 x_{i2}} \dots e^{\beta_k x_{ik}},$$

so that $h_i(t) \propto e^{\beta_1 x_{i1}} e^{\beta_2 x_{i2}} \dots e^{\beta_k x_{ik}}$. Thus increasing x_j by one changes the hazard function by a factor of e^{β_j} . This is the *hazard ratio*. For example, the hazard ratio for x_1 is

$$\frac{h_0(t)e^{\beta_1(x_1+1)} e^{\beta_2 x_2} \dots e^{\beta_k x_k}}{h_0(t)e^{\beta_1 x_1} e^{\beta_2 x_2} \dots e^{\beta_k x_k}} = e^{\beta_1},$$

since $e^{\beta_1(x_1+1)} = e^{\beta_1 x_1} e^{\beta_1}$.

Parametric Proportional Hazards Models

AFT models with a Weibull distribution (or exponential, which is a special case of the Weibull distribution) are also proportional hazards models. Consider the AFT model,

$$\log T_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \sigma \epsilon_i,$$

and the proportional hazards model

$$h_i(t) = h_0(t) \exp(\beta_1^* x_{i1} + \beta_2^* x_{i2} + \dots + \beta_k^* x_{ik}),$$

where in both cases T_i has a *Weibull* distribution. It can be shown that the models are equivalent with

$$\beta_j^* = -\beta_j / \sigma.$$

The hazard ratios are $e^{\beta_j^*}$.

An AFT model with a Weibull distribution is the *only* AFT model that is also a proportional hazards model. Other proportional hazards models exist, but none of them are AFT models.

Example: We can estimate a Weibull proportional hazards model for the **leukemia** data using **survreg** as follows.

```
m <- survreg(Surv(time, status == "dead") ~ x, dist = "weibull", data = leukemia)
summary(m)
```

Call:

```
survreg(formula = Surv(time, status == "dead") ~ x, data = leukemia,
        dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	4.109	0.300	13.70	<2e-16
xNonmaintained	-0.929	0.383	-2.43	0.015
Log(scale)	-0.235	0.178	-1.32	0.188

Scale= 0.791

Weibull distribution

Loglik(model)= -80.5 Loglik(intercept only)= -83.2

Chisq= 5.31 on 1 degrees of freedom, p= 0.021

Number of Newton-Raphson Iterations: 5

n= 23

The estimated hazard ratio is $e^{\hat{\beta}_1^*}$ where $\hat{\beta}_1^* \approx 0.929/0.791 \approx 1.175$ so $e^{\hat{\beta}_1^*} \approx 3.238$. Thus

$$\frac{h_n(t)}{h_y(t)} = e^{\beta_1^*} \Leftrightarrow h_n(t) = e^{\beta_1^*} h_y(t),$$

where we estimate the hazard ratio $e^{\beta_1^*}$ to be 3.238. This conversion can be done using the `ConvertWeibull` function from the `SurvRegCensCov` package.

```
library(SurvRegCensCov)
ConvertWeibull(m)
```

\$vars

	Estimate	SE
lambda	0.005544	0.005698
gamma	1.264295	0.225328
xNonmaintained	1.174962	0.523035

\$HR

	HR	LB	UB
xNonmaintained	3.238	1.162	9.026

\$ETR

	ETR	LB	UB
xNonmaintained	0.3948	0.1866	0.8356

Another approach is to use `dist = "weibullPH"` with `flexsurvreg` which uses a different parameterization of the Weibull distribution so that applying the exponential function to the parameters gives hazard ratios.

```
m <- flexsurvreg(Surv(time, status == "dead") ~ x, dist = "weibullPH", data = leukemia)
print(m)
```

Call:

```
flexsurvreg(formula = Surv(time, status == "dead") ~ x, data = leukemia,
            dist = "weibullPH")
```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)	L95%	U95%
--	------	------	-----	------	------	----	----------	------	------

shape	NA	1.26430	0.89156	1.79286	0.22532	NA	NA	NA
scale	NA	0.00554	0.00074	0.04156	0.00570	NA	NA	NA
xNonmaintained	0.52174	1.17496	0.14984	2.20009	0.52303	3.23802	1.16165	9.02580

N = 23, Events: 18, Censored: 5
 Total time at risk: 678
 Log-likelihood = -80.52, df = 3
 AIC = 167

The proportionality can be seen when plotting the hazard functions.

```

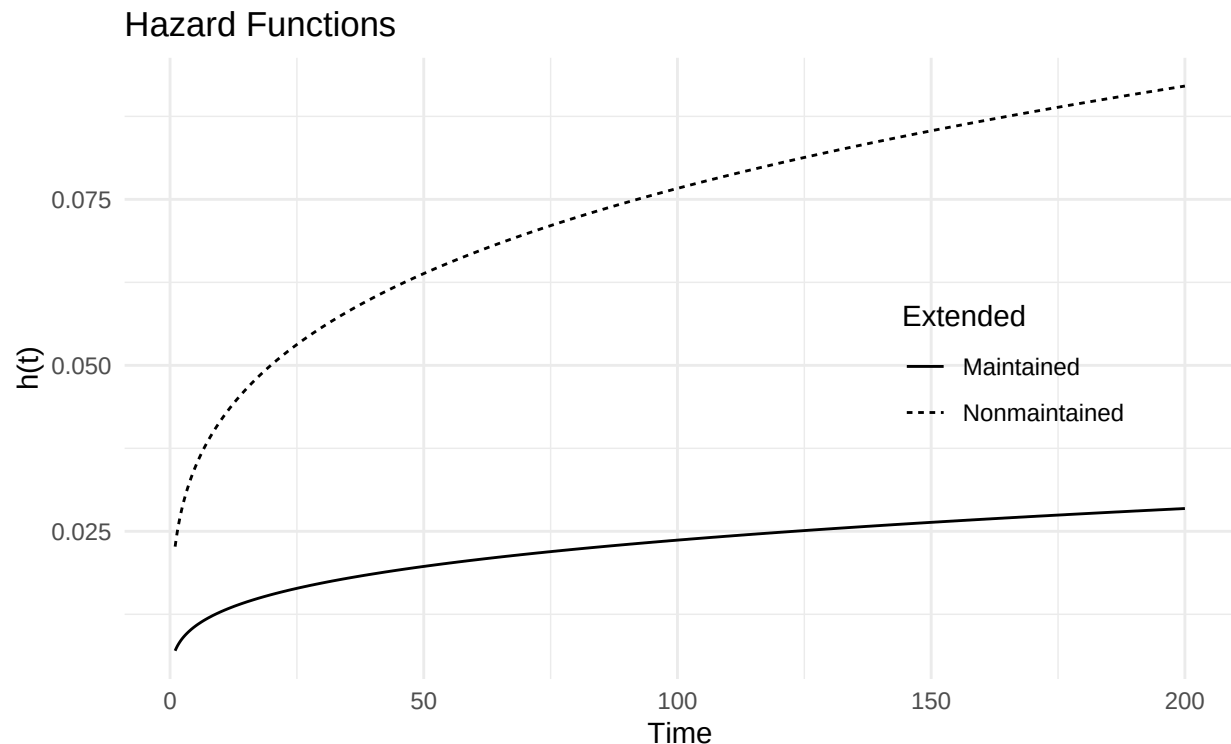
d <- data.frame(x = c("Maintained", "Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "hazard", tidy = TRUE)

```

```

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "h(t)", linetype = "Extended", title = "Hazard Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.8, 0.5))
plot(p)

```



Example: Consider a Weibull proportional hazards model for the motors data.

```

m <- flexsurvreg(Surv(time, cens) ~ temp, data = MASS::motors, dist = "weibullPH")
print(m)

```

Call:

```

flexsurvreg(formula = Surv(time, cens) ~ temp, data = MASS::motors,
  dist = "weibullPH")

```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)	L95%	U95%
--	------	------	-----	------	------	----	----------	------	------

shape	NA	2.99e+00	1.97e+00	4.55e+00	6.40e-01	NA	NA	NA
scale	NA	6.34e-22	1.79e-30	2.24e-13	6.37e-21	NA	NA	NA
temp	1.82e+02	1.36e-01	8.04e-02	1.91e-01	2.81e-02	1.15e+00	1.08e+00	1.21e+00

N = 40, Events: 17, Censored: 23
 Total time at risk: 140654
 Log-likelihood = -147.4, df = 3
 AIC = 300.7

Here we have that

$$h_{x+1}(t) = e^{\beta_1^*} h_x(t),$$

where $h_x(t)$ and $h_{x+1}(t)$ represent the hazard functions at temperatures of x and $x + 1$, respectively. The estimated hazard ratio is $e^{\hat{\beta}_1^*} = 1.15$.

```
d <- summary(m, newdata = data.frame(temp = seq(110, 150, by = 10)),
  t = seq(0, 8000, length = 1000), type = "hazard", tidy = TRUE, ci = FALSE)

p <- ggplot(d, aes(x = time, y = est, color = factor(temp))) +
  geom_line() + theme_minimal() +
  theme(legend.position = "inside", legend.position.inside = c(0.2, 0.6)) +
  labs(x = "Hours", y = "h(t)", color = "Temperature", title = "Hazard Functions")
plot(p)
```

